

Supersymmetry Dataset (DSc Project Report)

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Abstract

This project applies data science and machine learning techniques to classify high-energy particle collision events using the Supersymmetry (SUSY) dataset. Through preprocessing, feature selection, and dimensionality reduction (PCA and JL matrices), we optimized the dataset for efficient model training. Logistic Regression, Random Forest, SVM, and Perceptron models were evaluated, achieving reduced training times with minimal accuracy trade-offs.

1. Introduction

High-energy particle colliders have long been pivotal in advancing our understanding of fundamental physics, serving as a fertile ground for the discovery of exotic particles. These discoveries, such as the landmark identification of the Higgs boson, offer profound insights into the nature of matter and the fundamental forces governing the universe. At the heart of these discoveries lies the challenge of distinguishing rare signals indicative of new particles from the overwhelming background noise in experimental data.

Machine learning has emerged as a powerful tool for addressing this challenge, enabling researchers to develop sophisticated signal versus background classification models. In the context of high-energy physics, the problem can often be framed as a binary classification task: determining whether a specific kinematic configuration of colliding particles results in the production of a supersymmetric particle (labeled as 1) or constitutes a background process (labeled as 0).

Conducting such experiments is inherently resource-intensive, requiring state-of-the-art particle accelerators, skilled personnel, and significant financial investments. Moreover, the combinatorial complexity of testing all possible kinematic configurations makes direct experimentation tedious to conduct. By leveraging machine learning and analytical approaches, researchers aim to predict experimental outcomes based on patterns and trends observed in past data. This approach not only reduces the need for exhaustive testing but also accelerates the pace of discovery in particle physics.

This report explores the application of machine learning methods to classify high-energy collision events, with the goal of identifying supersymmetric events and advancing our understanding of exotic particle physics.

2. Dataset Description

The **Supersymmetry (SUSY) dataset** is a prominent dataset in the field of high-energy physics, created to support binary classification tasks for particle collision events. Derived from experiments conducted using particle accelerators, the dataset provides a rich resource for analyzing the kinematic properties of particle interactions. Its primary application is in distinguishing between background processes and supersymmetric events, making it an excellent tool for machine learning applications in physics.

This dataset is rooted in the domain of particle physics and consists of 5 million instances, offering a substantial sample size for training and evaluation. Each instance resides in an 18-dimensional feature space, capturing detailed insights into the kinematic behavior of particles during collisions. The target variable is binary, classifying collision events into one of two categories: a background process, where no new particle is generated (labeled as 0), or a supersymmetric event, where an exotic particle is produced (labeled as 1). The number of 0's and 1's in the dataset are almost identical, with the mean of the target variable approximately 0.46. This balanced class distribution eliminates the need for techniques to handle class imbalance, simplifying the process of model evaluation as demonstrated in pie chart 1. Also, there were no missing values in the dataset.

The dataset's features are continuous and represent the kinematic properties of particles involved in the collisions. The features of the dataset are explain in the table 1.

3. Methodology

3.1. Data Preprocessing

We plotted Histograms for each feature and found that Lepton 1 phi, Lepton 2 phi, and Missing Energy phi are evenly distributed across their respective ranges, indicating uniform variability. Additionally, the features Lepton

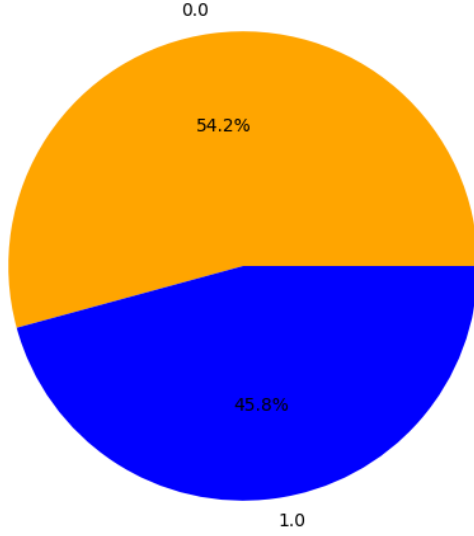


Figure 1. Class Distributions

1 eta and Lepton 2 eta exhibited highly similar distributions, suggesting potential high correlation between them. Identifying such patterns helped to determine which features might introduce redundancy during modeling.

The preprocessing phase involved several methods to prepare the SUSY dataset for analysis and model training. Firstly, the dataset was free of missing values, this eliminated the need for additional preprocessing steps related to handling missing data.

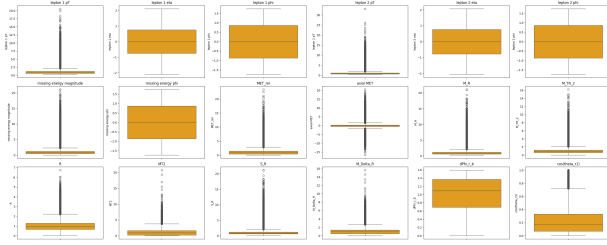


Figure 2. Box Plots of raw features

Boxplot analysis revealed the presence of significant outliers across multiple features. To address this, the Interquartile Range (IQR) method was employed. This method identified data points as outliers if they were outside 1.5 times the IQR from the first quartile (Q1) and third quartile (Q3). Removing these outliers significantly improved data consistency and reduced the dataset size from 5 million instances to 3,689,288 instances. This step ensured that the dataset was free of extreme values that could otherwise bias the training process.

Lastly, all feature values were already within a range of +2.5 to -1.5 after outlier removal. This uniform range across features made scaling unnecessary, stream-

Table 1. Dataset Features

Feature	Meaning
lepton 1 pT	Transverse momentum of the first lepton
lepton 1 eta	Pseudorapidity of the first lepton
lepton 1 phi	Azimuthal angle of the first lepton
lepton 2 pT	Transverse momentum of the second lepton
lepton 2 eta	Pseudorapidity of the second lepton
lepton 2 phi	Azimuthal angle of the second lepton
missing energy magnitude	Magnitude of missing transverse energy
missing energy phi	Azimuthal angle of missing transverse energy
MET_rel	Relative missing transverse energy
axial MET	Axial missing transverse energy
M_R	Mass scale related to the system of particles
M_TR_2	Transverse mass variable
R	Ratio of two transverse mass variables
MT2	Minimal transverse mass variable
S_R	Scalar sum of certain transverse momentum variables
M_Delta_R	Mass difference between particles
dPhi_r_b	Azimuthal angle difference between certain particle systems
cos(theta_r1)	Cosine of the polar angle of a particle system

lining the preprocessing workflow while maintaining the dataset's integrity for modeling.

3.2. Hypothesis Testing

To refine the feature set and improve the predictive power of the model, hypothesis testing was conducted using statistical methods to evaluate feature relevance and redundancy. The following steps outline the process:

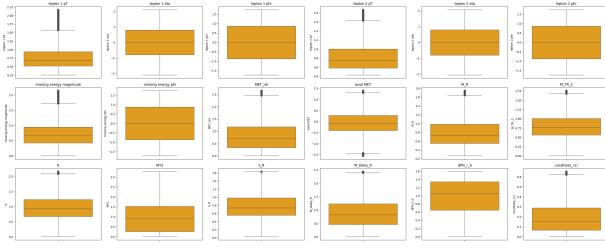


Figure 3. Box Plots after outlier removal

3.2.1 Analysis of Variance (ANOVA)

An ANOVA test was performed to assess whether each feature varies significantly between the two classes (0 and 1). The null hypothesis for this test was that there is no significant variation in the feature values across the classes. Features with an ANOVA score less than 3000 were considered to have insufficient predictive power and were removed from the dataset. This step ensured that only features contributing significantly to class differentiation were retained.

3.2.2 Spearman's Correlation Hypothesis Testing

To identify and reduce feature redundancy, hypothesis tests for Spearman's correlation coefficient were conducted. Spearman's correlation measures monotonic relationships (both linear and nonlinear) between features. The tests were conducted with a critical p -value of 0.05 to determine the statistical significance of the correlation.

- **Null Hypothesis (H_0):** No monotonic correlation exists between the features; Spearman's coefficient = 0.
- **Alternate Hypothesis (H_a):** A monotonic correlation exists between the features; Spearman's coefficient $\neq 0$.

Findings and Feature Redundancy Reduction

- **M_R and S_R :** These features showed a very high Spearman coefficient of 0.91, indicating a strong monotonic relationship. Between the two, M_R exhibited a higher Spearman coefficient with the label (0.28 compared to 0.26). Therefore, S_R was dropped to reduce redundancy.
- **$MT2$ and M_R :** These features also showed a very high Spearman coefficient of 0.92. Between the two, M_R had a higher Spearman coefficient with the label (0.14 compared to 0.04). As a result, $MT2$ was dropped to reduce redundancy.

By leveraging both ANOVA and Spearman's correlation tests, the feature set was refined to retain features with high predictive power while minimizing redundancy. Finally we reduced the dataset from 18 to

10 features namely: **lepton 1 pT**, **lepton 2 pT**, **missing energy magnitude**, **MET_rel**, **M_R**, **M_TR_2**, **R**, **M_delta_R**, **dPhi_r_b**, **cos(theta_r1)**.

3.3. Building Models

We trained Logistic Regression, Random Forest, Support Vector Machine(SVM) and Perceptron models and mentioned the accuracy, recall, precision, F1-score of each model in Table 2.

Table 2. Final Accuracy, Precision, Recall, and Macro F1 Scores of models on raw dataset

Model	Accuracy	F1-Score	Training Time (s)
Logistic Regression	0.766	0.75	14.55
Random Forest	0.773	0.76	285.8
SVM	0.624	0.57	484.82
Perceptron	0.660	0.59	2.15

Then as expected the raw dataset takes a lot of time to run any model and especially Random Forest with the best accuracy of 77.3% takes 285.8 sec to run. So, now we use the selected 10 features and further applied dimensional reduction methods to improve the running time while preserving as much performance as possible.

So, we applied two dimensionality reduction methods PCA and JL matrices and observed their impact by comparing the model metrics.

3.3.1 PCA

It was observed that only 6 components were able to explain more than 95% of variance in the data as shown in fig 4, we transformed the data reduced the number of dimensions to 6 and retrained the models. The new model results are mentioned in Table 3

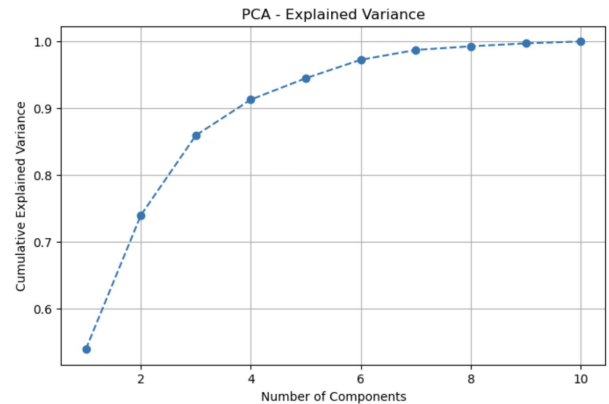


Figure 4. Number of features vs variance in data explained by them

3.3.2 JL Matrices

The data matrix D of size approximately 3.7×10 was transformed using a JL matrix, J , of size 10×5 . The JL

Table 3. Final Accuracy, Precision, Recall, and Macro F1 Scores of models on raw dataset

Model	Accuracy	F1-Score	Training Time (s)
Logistic Regression	0.743	0.73	1.41
Random Forest	0.753	0.74	145.66
SVM	0.662	0.54	182.95
Perceptron	0.550	0.55	1.56

matrix was constructed with its elements sampled from a normal distribution, $N(0, 1/\sqrt{10})$, to ensure the properties of the transformation.

The dimensionality reduction was performed by multiplying the data matrix D with the JL matrix J , resulting in a transformed data matrix D' , with a reduced size of approximately 3.7×5 . This reduced dataset was used to re-train the models and the new metrics are mentioned in the Table 4.

Table 4. Final Accuracy, Precision, Recall, and Macro F1 Scores of models on raw dataset

Model	Accuracy	F1-Score	Training Time (s)
Logistic Regression	0.751	0.74	2.07
Random Forest	0.7504	0.74	133.98
SVM	0.467	0.42	214.75
Perceptron	0.680	0.58	1.59

4. Challenges Faced

The challenges faced during the course of the project were as follows:

1. Most datasets that were publicly available on the internet were of very small size, with the number of instances and dimensions being very low. Selecting the final dataset required extensive searching, and we eventually arrived at a dataset that had an adequate number of instances (in the range of millions) and features so that the scaling methods could be applied effectively.
2. Initially, we tested feature correlations for feature selection using the Pearson correlation coefficient, which only captured linear relationships between features. However, we soon discovered that this approach produced inconsistent results, as even a simple monotonic relationship such as $y = x^3$ could not be captured. We shifted to using Spearman's correlation coefficient, which captured both linear and non-linear monotonic correlations, making the feature selection process more precise.
3. Training the models or performing large matrix multiplication operations sometimes took a significant amount of time and caused intensive device heating and high RAM usage. This was mitigated by limiting the hyperparameters to a specific range and making matrices sparse to reduce computational complexity.

5. Conclusion and Future Scope

In this project, we analyzed the Supersymmetry (SUSY) dataset to solve a binary classification problem involving particle physics collision events. Through data preprocessing, feature selection, and dimensionality reduction techniques, we optimized the dataset for effective training and testing. By employing four different models—Logistic Regression, Random Forests, SVM, and Perceptron—we evaluated their performance under various scenarios, including raw data, feature selection, and dimensionality reduction methods like PCA and JL matrices. These scaling methods worked well as they embed data into a lower dimensional space with only some loss of information, JL matrices preserve the distances between the points even in a lower dimensional space. This reduced dataset contained sufficient information to train a successful classifier which could perform decently on unseen data. Key observations included significant reductions in training time with minimal impact on accuracy and other performance metrics, demonstrating the effectiveness of the scaling methods that were discussed in the course.

- **Exploring Additional Models:** Future work can involve experimenting with a wider range of models for binary classification to assess their performance and identify the most suitable one for the dataset and problem context.
- **Hyperparameter Optimization:** Further testing can be done by varying hyperparameters more extensively, using techniques like grid search or randomized search to find optimal configurations for improved model performance.
- **Alternative Scaling and Dimensionality Reduction Methods:** Investigating other techniques for scaling and dimensionality reduction could help in enhancing computational efficiency and feature representation.

6. References

Dataset : [Supersymmetry \(SUSY\) Dataset](#)
All figures, tables and graphs are referenced from our code which can be found at the given Github [link](#)

7. Group Details

Group Name : Sigma Squared
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